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A Systematic Literature Review of Classical and Quantum Machine Learning Approaches for Mutual Fund Portfolio Optimization

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***Abstract*—Quantum Computing is a dynamic field which has evolved since its inception around 3 decades back, to the development of various powerful algorithms including Quantum Approximate Optimization Algorithm (QAOA) and Variational Quantum Eigensolver (VQE) to name a few. Today's quantum computers are not as powerful as the current breed of supercomputers. However, the existing breed of NISQ (Noisy Intermediate Scale Quantum) quantum computers have a significant potential to provide faster solutions to problems in various domains which are not just relevant for the present but also for the future such as Quantum Machine Learning in the Fintech domain. Portfolio Optimization (PO) is one such problem in research and the financial services industry at large. In principle, Quantum Computing, makes it possible to arrive at an optimal portfolio composition for a high return and low risk investment much faster than any existing supercomputer today. This review paper examines literature on classical and quantum machine learning approaches for Mutual Fund PO, analyzing 44 papers from 2003 to 2023. PO being a vast topic, we limit our scope to Mutual Funds and its driver that is the equity markets. We provide an overview to the types of problems, preferred approaches, their benchmarks, deduced conclusions, and research gaps as a comprehensive survey for diverse readers.**

***Keywords*—Quantum Computing, Mutual Fund, Portfolio Optimization, Quantum Machine Learning, quantum annealer**

I. Introduction

Apart from physics and computer science, from the perspective of Quantum applications, there are other domains where quantum computing (QC) research work is being carried out including Banking and Financial Institutions [1], Pharma, [2]Supply Chain Logistics [3] etc. Optimization problems are synonymous in every field, for example: from flight scheduling to deployment of transportation vehicles. At the heart of QC processing, lies quantum bits (qubits) which characterizes the true potential of the quantum computer. Despite the fact that fault tolerance in QC with near perfect logic qubits still needs to be accomplished, the technology has drawn attention with the advantages through its small-scale implementations.

In addition to the phenomenon of demonstrating multiple states called as superposition, another property of entanglement permits qubits to be linked, so that the state of one directly influences the state of others regardless of the distance. This capability enables quantum computers to concurrently process immense quantities of data, resulting in considerably superior computational speeds compared to conventional systems[4]. In spite of the inherent difficulties of maintaining quantum states and minimizing decoherence, substantial progress has been made. With breakthroughs like the Shor's algorithm for factoring and others paving the way for future research.

The current stage of quantum technology research for devices with 50–1000 qubits that are not sufficiently advanced to produce fault-tolerant computers is known as noisy intermediate-scale quantum computing, or NISQ. [5]These processors are noisy, susceptible to quantum decoherence, and unable to do continuous quantum error correction. Even though NISQ devices have limitations, they can still provide valuable results and improve our understanding of using quantum principles for computing. In short, these devices can perform calculations better than classical supercomputers, but the results may not always be exact because of the noise in these systems.

Grover's algorithm was the first of its kind to outperform the finest classical search algorithms. Its success makes it possible to perform a search operation much more quickly on a quantum computer, it has the potential to provide a solution for the current NP hard, complex financial problems [6][7], [8]. A key area where quantum computing could have a significant impact is portfolio optimization, and asset allocation [9] Portfolio optimization (PO) is a computationally complex activity that necessitates scanning through several prospective portfolios. Quantum algorithms can efficiently search for these combinations and find better risk and return management as an optimal solution in a fraction of the time. In the area of risk assessment and modelling, quantum computing will enable financial institutions to perform simulations and stress tests at a large scale.

The performance of Machine Learning (ML) techniques used in finance [10] could also be improved by quantum computing. Quantum Machine Learning (QML) PO algorithms which are an intersection of QC and ML techniques, process large datasets more efficiently, revealing hidden patterns and insights that traditional ML approaches may potentially not be able to identify [7]. This improves predictive models and enables informed investment

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decisions. This paper focuses on Mutual Fund (MF) because it has seen a rise in investment in the past years and a low rate of risk in comparison to the ever-fluctuating stock market industry [11]. Apart from the five significant Mutual Fund problems described in Table 1, Liquidity Management, Benchmark Tracking, Factor Investing and Constraints Compliance are some of the other problems. Thus, combining financial experience, statistical analysis, and computational power is crucial for efficient portfolio management in mutual funds to solve complex optimization problems.

TABLE I. DESCRIPTION OF TYPES OF PROBLEMS

Mutual Fund Portfolio Optimization	
Types of Problems	***Description***
Asset Allocation	Determining the best distribution of assets among various asset classes (such as equities, bonds etc.) to meet the investment goals of the fund. In this problem, risk and return must be balanced while taking past performance, anticipated returns, and correlations between asset classes into account.
Portfolio Diver-sification	The process of choosing a varied group of securities from among those offered by each asset class in order to lower unsystematic risk and enhance the portfolio's risk-return profile. Finding the ideal combination of securities that are not very correlated with one another is necessary for optimization.
Risk Manage-ment	It involves limiting different risk indicators, such as volatility, beta, or Value at Risk (VaR)[1], to make sure the portfolio's degree of risk is consistent with the risk tolerance and mandate of the fund.
Minimizing Transaction Costs	It is essential when rebalancing the portfolio or making other changes in order to protect investors' returns. Finding the most economical strategy to acquire or sell assets while taking market impact into account is the goal of this problem.
Tax Efficiency	In taxable mutual funds, this refers to managing the portfolio to reduce investor tax obligations and capital gains distributions.

Choosing the right benchmark is crucial in Mutual Fund selection, like the Treynor Index, Sharpe Ratio, etc. Although there is an existence of tools that tackle PO, which is non-convex by nature and utilizes the cost function (gradient descent methods etc.), conventional computing techniques find it hard to solve it because such problems lack the nature of possessing a simple mathematical structure [6].

II. RESEARCH METHODOLOGY

This section gives details of the PRISMA methodology that was followed to find results on a topic like QML in mutual fund portfolio optimization, source material from the past 20 years was selected and investigated. The references were traced back to Scopus, arXiv.org, and other academic search engines. Selected journals include Nature, Quantum Journal, Physical Review Research, Quantum Information Processing, etc. Fig. 1, is a pictorial representation of the approach adopted.

The search yielded papers focusing on stocks as the underlying data rather than mutual funds data. This alone is a significant gap in the existing literature. Thus, the reference works included are primarily those that concentrate on mutual funds or equities, particularly those that have benefited from QML and ML.

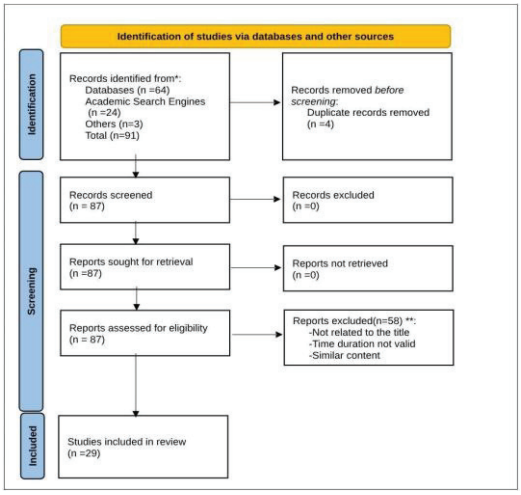

Fig. 1. The methodology followed for the structured literature review paper.

***** Exceptions were made to two papers, both of them being milestone papers with 400+ citations***
****Scopus as a major source in addition to arXiv, SSRN etc.***

From the chart for Fig. 2, it is evident that QML is a developing field, which is also recorded in [12]. Our statistics show an approx. 67.81% of the collection ranging in the year 2019 to 2023.

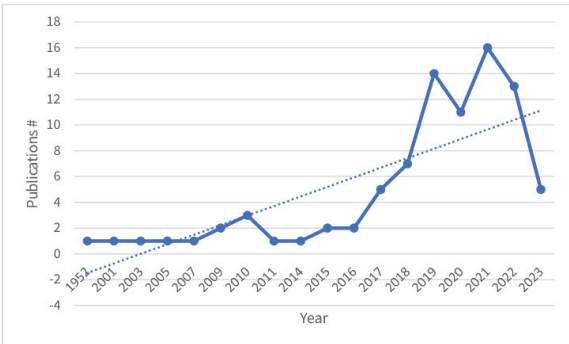

Fig. 2. The survey comprises of 87 papers and the chart depicts the trend in the publication year.

Based on the above and establishing a context for ML and QML in Portfolio Optimization, this research review aims to address the following specific research questions that will shape our inquiry.

RQ1: ***What is the prominent research works that have been carried out in classical and quantum machine learning to address an np-hard task like mutual fund portfolio optimization?***

RQ2: ***What are the critical research gaps in the work that has been carried out?***

III. LITERATURE REVIEW

For the past 20 years, mutual funds and capital markets have been very attractive research topics. In comparison to emerging economies, industrialized countries have a large body of literature on capital market research. The author affiliation of Table II, showcases USA as leader for this survey. From a practical perspective, humans lack the ability to choose a Mutual Fund or stock that will do well in the

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future. As a result, the utility of enhanced ML technologies like QML for financial goals should be explored.

TABLE II. PORTFOLIO OPTIMIZATION PROBLEMS IN THE WORLD

Distribution of Authors		
Reference Paper	***Country***	***Type***
Portfolio optimization [11]	Malaysia	Equity MF
Portfolio optimization with fuzzy risk and return rates [13]	Taiwan	Equity MF
An approach to portfolio optimization problems using reverse quantum annealing [14]	USA, UK	Fund of Funds
Portfolio Optimization using genertic algorithm [15]	South Korea	Index Funds
A simulation of Dynamic Portfolio Management Using Neural Network Predictive Modelling [16]	USA	Index Mutual Funds
A portfolio selection strategy using fuzzy goal programming [17]	Spain	Mutual Funds
A combined approach of machine learning and portfolio selection for financial trading decisions [18]	Brazil	Stocks
Quantum computing and quantum-inspired tensor networks for dynamic portfolio optimisation on real datasets [9]	Canada, Spain	Stocks
Quantum optimization use cases for Finance [6]	Canada, Spain	Stocks
Insights from a finance application using machine learning with classical versus quantum model approach [7]	Canada, UK, USA	Stocks
S&P 500 stock selection using a deep reinforcement learning agent [19]	Finland	Stocks
Can portfolios based on machine learning outperform conventional risk-based portfolios? Taking covariance misspecification into account [20]	India	Stocks
Machine learning based and financial indicators for Investement techniques in the Global stock. [21]	Korea	Stocks
Approaches to Finance using Deep learning [22]	USA	Stocks
A recurrent reinforcement lerning for a risk-retun portfolio for an adaptive trading system. [23]	USA	Stocks
Deep reinforcement learning with a model-based approach for dynamic portfolio optimization [24]	Singapore	Stocks
Using Simulated, Digital and Quanum Annealing for Strategic Portfolio Optimization. [25]	Germany, Netherlands	Stocks & ETFs

A. Classical Machine Learning Approaches for PO

In this work, [26] the authors highlight the ability of ML to self-learn, reduce dimensionality, build non-linear relationships and improve over time which also justifies its use in the sector of finance. Return forecasting, portfolio design, and risk modelling were the three most common use cases at the core of quantitative methods among the 67 papers analyzed by [27].

The statistical distribution in the pie chart, Fig. 3. indicates, construction was the second popular areas with 22 papers amounting to 33%.

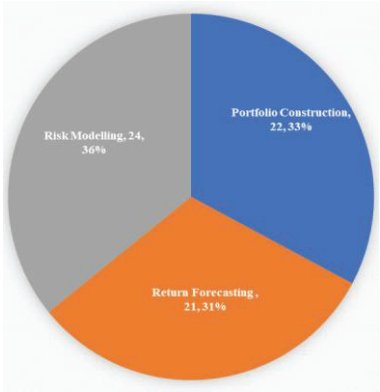

Fig. 3. Sub areas for quantitative finance that employ Machine Learning Algorithms

Portfolio optimization problems may be categorized as either convex or combinatorial problems of the following three forms: Active investment management having continuous weights (convex), Active investment management having binary weights (combinatorial), Passive investment management and Portfolio diversification (combinatorial), Formulation of Mixed-Binary optimization. [1], [28].

The discrete nature of the portfolio optimization problem is NP-complete due to its strong non-linearity overpowering the continuous mean-variance portfolio optimization problem in terms of complexity levels [14]. PO was also examined to be a combinatorial optimization problem and research established that it was NP-hard [4], [28]. From this we conclude that PO problems are known to be the hardest to solve.

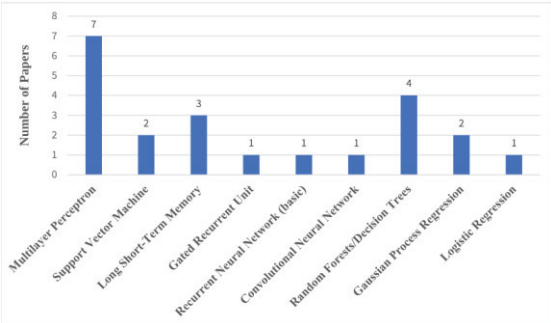

Fig. 4. Popular ML techniques for Portfolio construction from a pool of 22 papers.

Fig. 4. indicates that Multilayer perceptron was a popular ML method as far as portfolio construction is concerned in the collection of papers being considered, with Random forests and Decision Trees coming in second, followed by Long-Short-Term-Memory (LSTM), Support Vector Machine (SVM) and Gaussian Process Regression respectively. The same is also validated by [19], [26]. In this work, [13] the authors made use of a combination of fuzzy logic along with cluster analysis for Taiwan equity mutual funds.

The authors [11] used Equity Mutual Fund, variance, Treynor Index and rate of return. The authors [13] had chosen the 4 evaluation criteria as rates of return, standard deviation, Treynor Index, and Turnover rate. This research [17] utilized the covariance, expected return, variance, among others.

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Despite its positive characteristics, traditional ML approaches face time constraints, at times requiring days to run simulation-based epochs. They also have high costs due to their inability to consider risk calculations at various levels, including the curse of dimensionality, [4], [7], [10]. Despite being introduced in the 1950's the mean-variance Markowitz model is a fundamental foundation for portfolio optimization.
The quantum computing community is still researching its simplicity, relative correctness, and utility. Thus, paving the way to overcome the constraints of classical ML techniques.

B. *Quantum Machine Learning Approaches for PO*

Achieving quantum acceleration for the Mutual Fund PO could bring in significant benefits like taking into consideration the computation of macroeconomic conditions as one of the factors with real time results where accuracy matters [16]. Quantum-assisted ML algorithms offer a breakthrough in dynamic investment decisions as they can provide real-time solutions for complex market scenarios, especially in the domain of Mutual Funds PO.

An examination of the expressive abilities and efficacy of models: **classical vs. quantum** for datasets collected in the real world, is an important highlight to look forward to in the transition from real-world applications toward a quantum-ready future. Quantum Portfolio Optimization (QPO) is one application that has demonstrated experimental success, mapping the optimization work onto quantum technology [29]. Quantum algorithms have also been successfully implemented for portfolio optimization using a small-scale approach. A method to solve quantum linear system of equations introduced by Harrow, Hassidim and Lloyd (HHL) as an algorithm which forms the basis for several QML models. The NISQ-HHL algorithm is one such enhanced version of the former.

In the comparative study [14], as stated earlier in Table II, made use of Fund of Funds as the data source spanning across 1-year. The authors implemented Reverse Quantum Annealing for portfolio optimisation on the D-Wave Quantum annealer. The classical counterpart for this was the Genetic Algorithm approach. Another parallel implementation was carried out by [25] on a data set spanning across 5 years of adjusted closing price for ETF's and Stocks. This was once again carried out on the same hardware. The hybrid approach used was the Quadratic Unconstrained Binary Optimization (QUBO) technique and simulated annealing was chosen as the classical ML counterpart. The other examples of portfolio optimization are [6], [7], [9], [11], [13], [17] . Out of these, [9] utilized IBM-Q, whereas [7] made use of Ion-trap Quantum Computers.

The prominent classical benchmarks that have been used to evaluate the QML models for PO in two prominent papers were, the Genetic Algorithm Approach, Simulated Annealing, the Gekko solver (optimization based on python) and exhaustive solver.

IV. Discussion

The overall findings demonstrate not only the applicability of QML to solve the real-world np-hard financial PO combinatorial problem but also highlights how it overcomes the challenges of classical ML. One such challenge is the incorporation of the discrete constraints that form a part of the MF PO problem. The basic nature of Quantum Annealer is to find the best solution in a search space. Both the Quantum Annealer and the QUBO models are becoming the de facto standards for probing NP-complete and NP-hard issues in the context of quantum computing. It is possible to convert an NP-complete/hard problem into a QUBO representation.

After a careful study, some of the evident research gaps that were observed have been documented in Table III.

TABLE III. The Research Gaps

Table Research gaps that were found and important observations	
Reference	***Observation / Gap***
Future applications of quantum computing in the NISQ era [5]	The limited performance of NISQ technology makes it difficult to validate a quantum computer's output, prompting researchers to look for more effective ways to achieve higher quantum gate accuracy for future implementations.
Quantum Machine Learning for Finance [8]	Quantum linear-algebra techniques may not always be applicable for specific linear-algebra and financial use cases due to prerequisites and constraint accommodations and may become a bottleneck to achieve quantum speedups.
Dynamic Portfolio Management with Machine Learning [10]	No dynamic portfolio optimization framework can outperform the covariance model. ML/DL approaches require more research due to the curse of dimensionality and the DL architectures inability to improve performance of sample-based portfolios.

PO faces an increasing computational complexity, in terms of rebalancing portfolios in response to market fluctuations. The key challenges such as error correction, coherence time, and qubit stability must be addressed in order to maximize the utility of QC in the financial sector. With continued research and collaboration with quantum researchers and financial experts, breakthrough advancements can be expected in the financial industry, reshaping how finance is practiced, and open new avenues of innovation [1].

With numerous variables and conditions that need to be considered for a Mutual Fund PO problem, classical algorithms eventually end up at the local optima and offer a non-optimal solution.

Most implementations make use of hybrid techniques, which means that the algorithms combine classical and quantum computing. In the short term, this appears to be the most fruitful technique, as QCs have yet to acquire enough functional qubits to effectively optimize portfolios, unless a significant portion of the algorithm can be done on a traditional computer. Quantum Approximation Optimization Algorithm (QAOA) is one such quantum-based technique that offers a quick solution, with better quality solutions and answers to problems with vast parameters. PO is an area that should be researched more, based on early experimentation which has produced encouraging results.

A. *Critical Analysis of Significant Papers*

We provide a critical analysis of a few chosen milestone works as an addition to our SLR. Table IV is included, containing both the highest referenced papers (considered

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influential works) and significant papers that introduced any new PO algorithms or made improvements to already existing approaches.

TABLE IV. SIGNIFICANT PAPERS

Analysis	
Reference	***Main Contribution***
Quantum walk-based portfolio optimisation [4]	This paper proposes a quantum algorithm for portfolio optimization, targeted at NISQ-computers. Utilizing the Quantum Walk Optimization Algorithm, it helps in finding high-quality solutions.
Insights from a finance application using machine learning with classical versus quantum model approach [7]	This paper explores, how Quantum Circuit Born machines (QCBMs) exploite the probabilistic nature of quantum mechanics with an experimment on a near-term computer. Using Restricted Boltzmann machiness (RBMs), a direct comparison of classical and quantum models for unsupervised generative modeling has been performed.
Overview and prospect of Quantum computing for finance [29]	A balanced discussion on three computational approaches to financial problems focusing on optimization models in machine learning with neural networks. It discusses how quantum computing algorithms can be applied to finance to optimize portfolios, find arbitrage opportunities, and perform credit scoring. It also focuses on how quantum amplitude estimation lead to speed-ups for Monte Carlo sampling.

V. CONCLUSION AND FURTHER WORK

In conclusion, we have presented a comprehensive and systematic literature review of classical and quantum machine learning approaches for mutual fund optimization problems. We suggest that quantum machine learning will be crucial to optimization problems as technology advances. Real advantages in terms of solution quality, computing speed, or both can be expected to be found in specific uses cases. While many fields of research, including ML and optimization, stand to benefit from Quantum Computing, it is evident from the numbers that the specific application of QML to Mutual Fund PO issues have not received enough attention up to this point in time in spite of it having the potential.

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